

1. Administrative & Study Identification

Full Study Title: An Open-Label Extension Study of Voxelotor Administered Orally to Participants With Sickle Cell Disease Who Have Participated in Voxelotor Clinical Trials **Short Title/ Acronym:** Open-Label Extension of Voxelotor **Protocol Number:** GBT440-038 (Other ID: C5341023) **NCT Number:** NCT04188509 **Sponsor Name:** Global Blood Therapeutics Inc. (a wholly owned subsidiary of Pfizer) **Investigational Product:** Voxelotor (Oxbryta / GBT440) **Phase of Development:** Phase 3 **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the safety, tolerability, and Sickle Cell Disease (SCD)-related complications of long-term treatment with voxelotor in participants who have completed treatment in a prior Global Blood Therapeutics (GBT)-sponsored voxelotor clinical study. **Secondary Objectives:** To evaluate the long-term effect of voxelotor on clinical measures of anemia (e.g., hemoglobin levels), markers of hemolysis, and the annualized incidence rate of vaso-occlusive crises (VOCs).

2.2 Background & Rationale Sickle cell disease (SCD) is a genetic blood disorder characterized by abnormal hemoglobin (HbS) polymerization, leading to red blood cell sickling, chronic hemolytic anemia, and intermittent vaso-occlusive crises. Voxelotor is an orally bioavailable small molecule that binds to HbS, increasing its affinity for oxygen and inhibiting polymerization. This open-label extension (OLE) was designed to provide continued access to voxelotor and evaluate long-term safety, tolerability, and efficacy outcomes for patients who successfully completed previous GBT-sponsored trials. *(Note: The study was ultimately terminated early, and voxelotor was withdrawn from the market in late 2024 due to emerging clinical data indicating that the overall risk profile exceeded the observed benefits).*

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Open-label extension) **Blinding:** Open-label (Masking not used) **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: 162 participants **Number of Sites:** ~70 clinical sites globally **Estimated Study**

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female participant with SCD, aged ≥ 4 years. 2. Participated and received study drug in a GBT-sponsored voxelotor clinical study (Note: Participants who discontinued due to an AE but remained on study may be eligible if the AE poses no risk for retreatment; Participants who discontinued Study GBT440-032 as the result of an abnormal transcranial Doppler (TCD) flow velocity assessment ≥ 200 cm/sec are eligible). 3. Participant has provided written consent/assent (for pediatric participants, both the consent of the participant's legal representative or legal guardian and the participant's assent [where applicable] must be obtained).

4.2 Exclusion Criteria 1. Female participant who is breastfeeding or pregnant. 2. Participant withdrew consent from a GBT-sponsored voxelotor clinical study. 3. Known hypersensitivity to voxelotor or any other components of the study drug. 4. Use of St. John's wort, sensitive cytochrome P450 (CYP) 3A4 substrates with a narrow therapeutic index, or moderate/strong CYP3A4 inducers within 30 days of Day 1.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Participants aged ≥ 12 years and/or ≥ 40 kg receive a voxelotor dose of 1500 mg once daily (QD). Participants aged < 12 years and < 40 kg receive weight-based dosing to provide exposure corresponding to the adult dose. **Route of Administration:** Oral (administered as tablets, dispersible tablets, or powder for oral suspension/stick packs). **Duration of Treatment:** Long-term continuous daily treatment until the study termination.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care disease-modifying therapies for SCD (e.g., hydroxyurea, L-glutamine), provided doses are stable. **Prohibited:** Strong CYP3A4 inducers (e.g., St. John's wort) and sensitive CYP3A4 substrates with a narrow therapeutic index.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -14 to 0 Informed Consent, Medical History, Verification of prior GBT study completion, Pregnancy test
Baseline	Day 0	Enrollment, Weight Measurement (for pediatric dose calculation), First OLE Dose, Baseline Labs	Interim
Q12W	$\pm$ 7	Safety Labs, Weight	

assessment for dose adjustments, Efficacy Assessment, AE monitoring | | **End of Study** | Nov 2024 | Final Vital Signs, Exit Interview, IP Accountability (Study Terminated early) |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: 1. Number of Participants With Treatment Emergent Adverse Events (AEs). 2. Number of Participants With SAEs. 3. Number of Participants With Sickle Cell Disease (SCD) Related TEAEs and SAEs. **Measurement Method:** Clinical assessments, physical examinations, and continuous tracking of AE/SAE occurrences over the study duration.

7.2 Secondary & Exploratory Endpoints * Change from baseline in hemoglobin (Hb) concentration. * Change from baseline in markers of hemolysis (e.g., reticulocytes, unconjugated bilirubin, lactate dehydrogenase). * Annualized incidence rate of vaso-occlusive crises (VOCs).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via patient interviews, site visits, and scheduled clinical and laboratory assessments. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent safety monitoring evaluated the ongoing risk/benefit ratio. The accumulation of emerging clinical data indicating that the risk profile exceeded observed benefits ultimately led to the early termination of this trial and market withdrawal of the drug.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Analysis Set (all enrolled participants who received at least one dose of the study drug in the OLE). **Sample Size Justification:** Target $N = 162$ was based on the anticipated rollover rate of eligible participants from previous phase voxelotor trials, rather than a specific power calculation, given the open-label, single-arm extension design. **Handling of Missing Data:** Missing data in safety evaluations were generally not imputed.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits to ensure safety protocol adherence and proper weight-based dose

adjustments for pediatric participants. **Audit Trail:** eCRF data entry locking and data clarification forms via standard sponsor quality audit mechanisms.

11. Study Identifiers & Governance

Primary ID: GBT440-038 **Registry Number:** NCT04188509 **Sponsor/Lead Organization:** Global Blood Therapeutics Inc. (Pfizer) **Study Title:** Open-Label Extension of Voxelotor **Official Regulatory Title:** An Open-Label Extension Study of Voxelotor Administered Orally to Participants With Sickle Cell Disease Who Have Participated in Voxelotor Clinical Trials

12. Clinical Framework (Sickle Cell Disease Specific)

Primary Condition: Sickle Cell Disease (SCD) **Diagnosis Threshold:** Confirmed SCD diagnosis and completion of prior voxelotor clinical trial **Medical Methodology:** Hemoglobin S (HbS) polymerization inhibition **Optimization Target:** Improve Hb affinity for oxygen, decrease blood viscosity, and reduce chronic hemolysis

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for SCD supplemented with daily voxelotor. **Education Protocol (HCP):** Training on accurate pediatric weight-based dosing and dose adjustment thresholds. **Education Protocol (Patient):** Instructions for oral administration (e.g., swallowing tablets whole or proper preparation of dispersible tablets/powder). **Quality Audit Mechanism:** Ongoing safety signal detection, AE tracking, and patient adherence monitoring.

14. Observation & Follow-Up Schedule

Total Duration: Originally intended for long-term follow-up; terminated prematurely in November 2024. **Onsite Visit Cadence:** Generally every 12 to 24 weeks (Q12W - Q24W) for long-term safety extensions. **Remote Monitoring Frequency:** Routine interval outreach for AE reporting. **Checklist Review Interval:** Every 12 weeks for dose-adjustment evaluations.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--	
Principal Investigator	[Insert Name]	____	07-Apr-2026					

Clinical Operations Lead | [Insert Name] | ____ | 07-Apr-2026 | |
Lead Statistician | [Insert Name] | ____ | 07-Apr-2026 |